# Markdown Studio — Feature Demo

Open this file and run **Markdown Studio: Open Secure Preview** (`Cmd+Shift+P`).

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## 1. Markdown Rendering

**Bold**, *italic*, ~~strikethrough~~, `inline code`

> Blockquotes work too.

| Feature | Status |
|---|---|
| Markdown | ✅ |
| Mermaid | ✅ |
| PlantUML | ✅ |
| SVG | ✅ |
| PDF Export | ✅ |

1. Ordered list item
2. Another item
3. Third item

- Unordered item
- Another item

- Item

- 日本語

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## 2. Mermaid Diagrams

### Markdown Studio Architecture

### Extension Activation Flow

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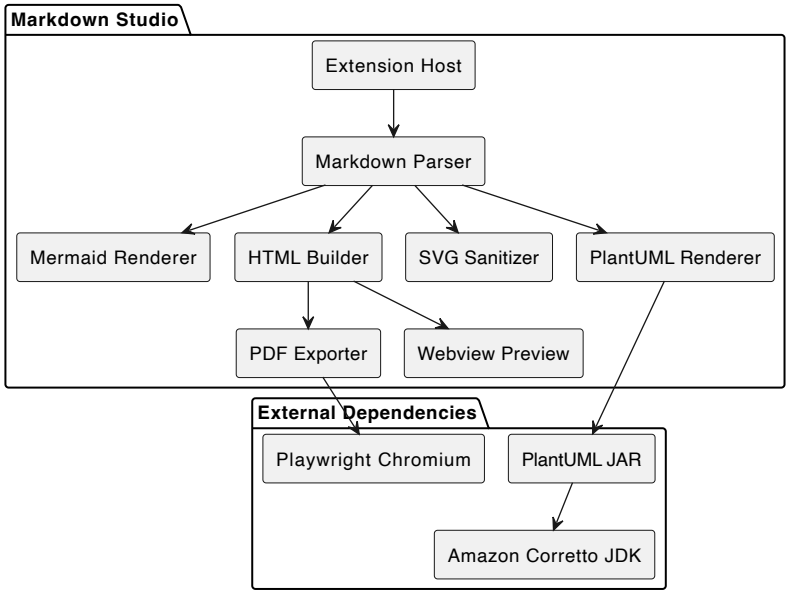
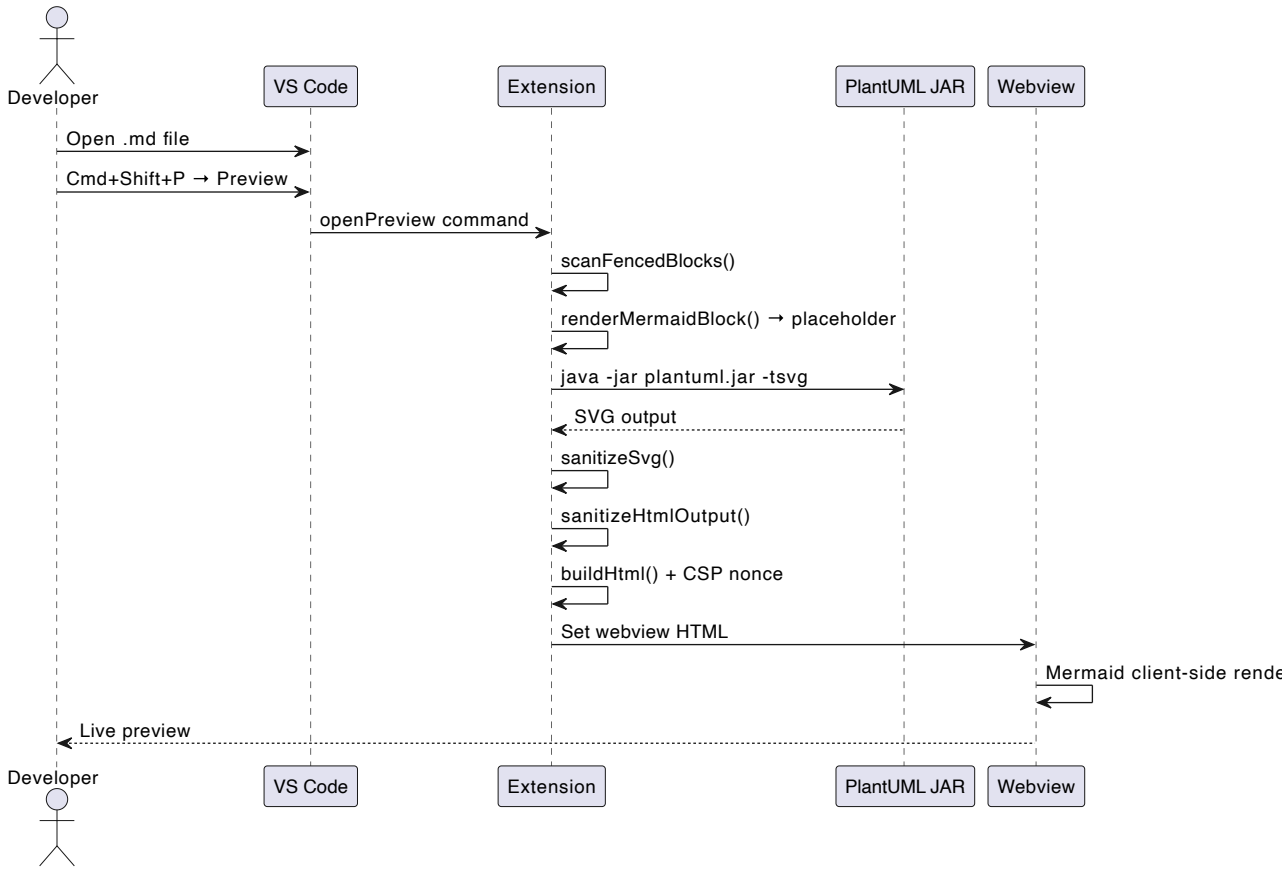
## 3. PlantUML Diagrams

Rendered locally via bundled JAR. No remote server.

### Extension Component Diagram



**Document Processing Sequence**

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## 4. Inline SVG (Sanitized)

Dangerous elements (`<script>`, `<foreignObject>`, event handlers) are stripped automatically.

Markdown Studio Pipeline

Parse markdown-it + scanner

Render Mermaid · PlantUML · SVG

Export Playwright → PDF

🔒 CSP + Nonce

🛡️ HTML Sanitize

💻 Local-First

Diagrams Syntax Highlighting Theme Aware

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## 5. Syntax Highlighting

```
import * as vscode from 'vscode';

export function activate(context: vscode.ExtensionContext): void {
  const depManager = new DependencyManager();
```

```
  const status = await depManager.ensureAll(context);

  context.subscriptions.push(
    vscode.commands.registerCommand('markdownStudio.openPreview', async () => {
      await openPreviewCommand(context);
    }),
    vscode.commands.registerCommand('markdownStudio.exportPdf', async () => {
      await exportPdfCommand(context);
    })
  );
}
```

```
{
  "markdownStudio.plantuml.mode": "bundled-jar",
  "markdownStudio.java.path": "java",
  "markdownStudio.export.pageFormat": "A4",
  "markdownStudio.security.blockExternalLinks": true
}
```

```
# PlantUML rendering is also useful for Python projects
from dataclasses import dataclass

@dataclass
class DiagramConfig:
    mode: str = "bundled-jar"
    java_path: str = "java"
    timeout_ms: int = 15000
```

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## 6. Security Model

- ✅ No external API calls
- ✅ No SaaS dependency or CDN assets
- ✅ Restrictive CSP with random nonce
- ✅ HTML sanitization before rendering
- ✅ SVG sanitization strips scripts, event handlers, foreign objects
- ✅ External links and images blocked by default

### External image (blocked by policy)

External image blocked by policy.

The image above is replaced with a policy notice in the preview.

### External link (blocked by policy)

External link — blocked by default ✋

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## 7. Theme Adaptability

Switch between **light** and **dark** mode in VS Code (`Cmd+K` `Cmd+T`) to see how the preview adapts:

- Mermaid diagrams automatically switch between light and dark themes
- SVG elements use colors chosen for visibility in both themes
- Code blocks use theme-aware syntax highlighting
- PlantUML output receives CSS overrides for dark mode

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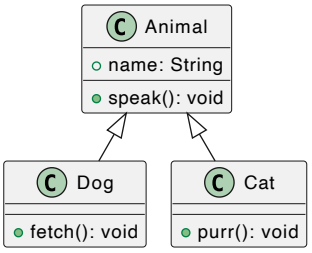
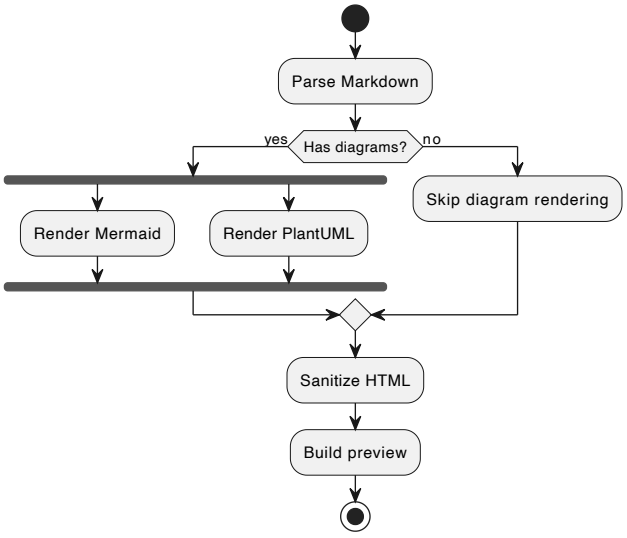
## 8. Diagram Type Catalog

All diagram types below are verified to render correctly with the bundled PlantUML + Smetana engine (no Graphviz required).

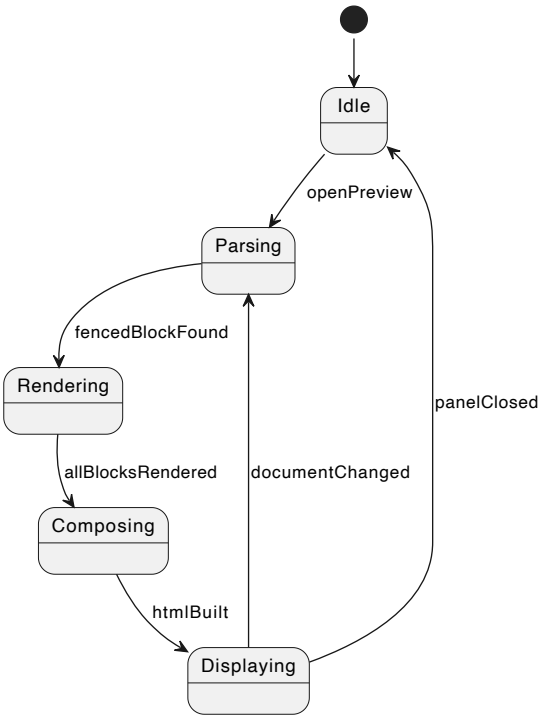
Class Diagram



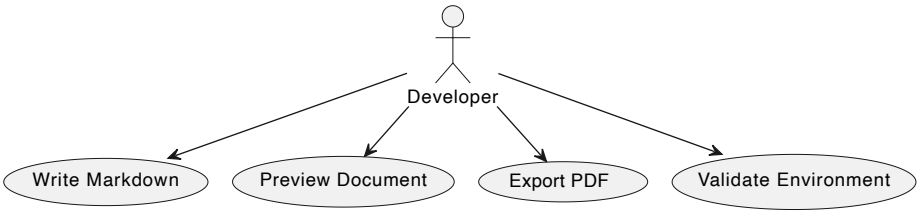

Activity Diagram



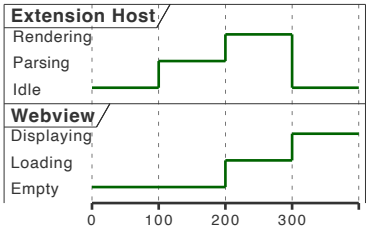

State Diagram



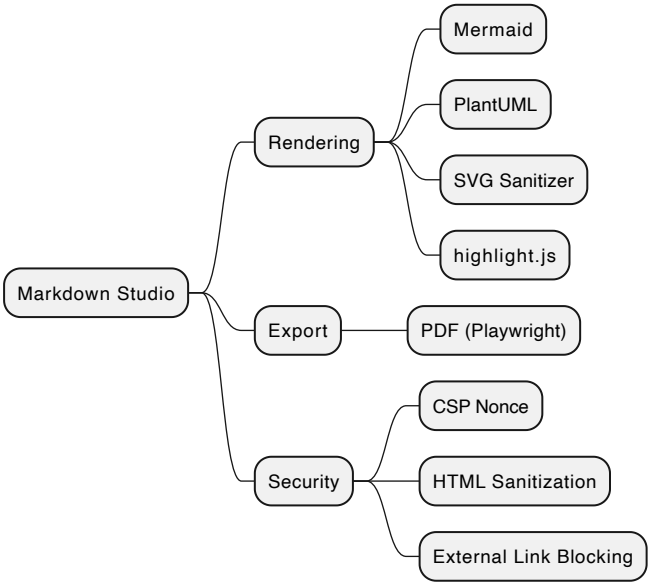

Use Case Diagram



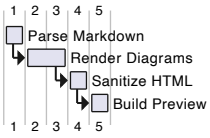

Timing Diagram



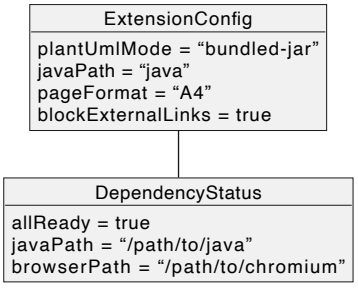

Mind Map




Gantt Chart




Object Diagram




Mermaid: Pie Chart

Mermaid: Git Graph

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## 9. PDF Export

This entire document can be exported to PDF:

1. Open this file in VS Code
2. `Cmd+Shift+P` → **Markdown Studio: Export PDF**
3. A `demo.pdf` will be generated next to this file

The PDF uses the same HTML pipeline as the preview.

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*Markdown Studio v0.2.0*